Exam: Electric and Magnetic Fields

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Instructions:

- Time: 45 minutes. Allowed aids: calculator. The formulas you need are given below.
- Show your work: write down the formula, insert values with units, and give the result with a sensible number of significant figures.
- Draw a sketch wherever it helps.
- Write your name on this cover sheet only. Write your solutions in the space below each question. If you need more room, use the last page.

Formulas

Electric field: $E = \frac{F}{q}$ Field of a point charge: $E = k\,\frac{Q}{r^2}$

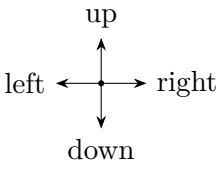
Field of a long, straight wire: $B = \frac{\mu_0}{2\pi}\,\frac{I}{r}$ Weight: $F_G = mg$

Newton's second law: $F = m\,a$

Constants

$k = \frac{1}{4\pi\varepsilon_0} = 8.99\cdot10^{9}\,\mathrm{N\,m^2/C^2}$, $\mu_0 = 4\pi\cdot10^{-7}\,\mathrm{T\,m/A}$, $e = 1.602\cdot10^{-19}\,\mathrm{C}$, $m_e = 9.109\cdot10^{-31}\,\mathrm{kg}$, $g = 9.81\,\mathrm{m/s^2}$

Directions



All directions refer to the page held upright in front of you: *up*, *down*, *left* and *right* as drawn on the page; gravity points *down*.

⊙ out of the page (towards you) ⊗ into the page (away from you)

Question:	1	2	3	4	5	6	7	Total
Points:	9	7	6	5	4	5	7	43
Score:								

Total solution time Mr. Spreder: 65 min ☺

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Do not write on this sheet: it is removed before marking.

1. Short questions. 5 min

(a) (3 points) A test charge $q = +2.0\,\text{nC}$ placed at point P experiences a force of $6.0 \cdot 10^{-5}\,\text{N}$ to the right. (i) What is the electric field at P (magnitude and direction)? (ii) What force would a charge of $-5.0\,\text{nC}$ experience at P? (iii) What is the field at P after the test charge has been removed?

i) $E = \frac{F}{q} = \frac{6 \cdot 10^{-5}\,N}{2 \cdot 10^{-9}\,C} = 3 \cdot 10^{4}\,\frac{N}{C}$ to the right

ii) $F = E \cdot q = 3 \cdot 10^{4}\,\frac{N}{C} \cdot 5 \cdot 10^{-9}\,C = 1.5 \cdot 10^{-4}\,N$ to the left

iii) The field remains unchanged as the test charge is not its source $\Rightarrow 3 \cdot 10^{4}\,\frac{N}{C}$

(b) (3 points) Convert: (i) $0.45\,\text{mT}$ into µT, (ii) $3.2 \cdot 10^{5}\,\text{N/C}$ into kN/C, (iii) $1.2 \cdot 10^{4}\,\text{nC}$ into mC.

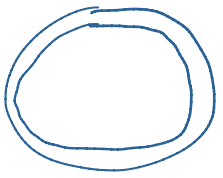
4 min

i) $0.45\,mT = 0.45 \cdot 10^{3}\,\mu T = 4.5 \cdot 10^{2}\,\mu T = 450\,\mu T$

ii) $3.2 \cdot 10^{5}\,\frac{N}{C} = 3.2 \cdot 10^{2}\,\frac{kN}{C}$

iii) $1.2 \cdot 10^{4}\,nC = 1.2 \cdot 10^{-2}\,mC$

5 min (c) (3 points) A loop of wire that carries a current acts as a magnet (an electromagnet). What happens to its magnetic field if (i) the current is increased, (ii) the current is reversed? (iii) Name one advantage of such an electromagnet compared to a permanent magnet.



background

long, straight wire: $B = \frac{\mu_0}{2\pi}\frac{I}{r}$

loop: different equation, but same dependence on I (superposition principle applied on moving charges)

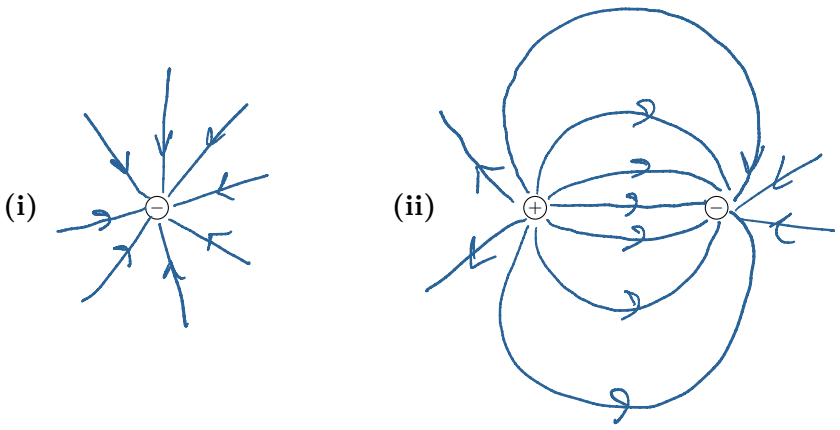
i) the magnetic field is increased $B \sim I$

ii) the magnetic field is reversed (right-hand rule)

iii) the field can be controlled by the current

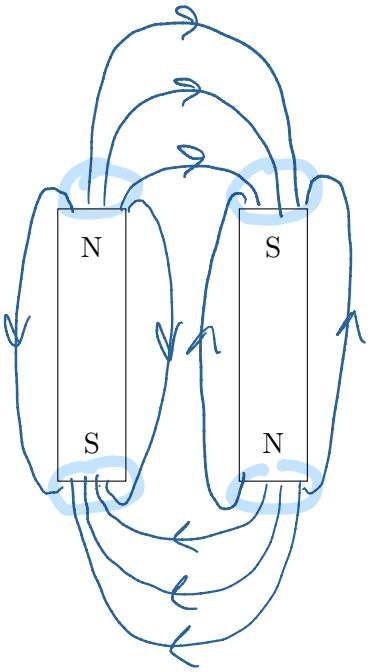
2. Field lines. 2 min

(a) (4 points) Sketch the electric field lines (with direction) of (i) a single *negative* point charge and (ii) a positive and a negative charge of equal magnitude.



(b) (3 points) Two bar magnets lie side by side, with opposite orientation (see sketch). Sketch the magnetic field lines (with direction). Where is the field strongest?

2 min

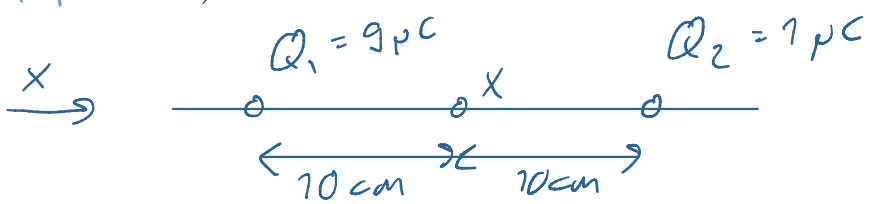


The field is strongest close to the poles (marked regions)

3. Two positive charges lie on a line: $Q_1 = +9.0\,\mu C$ at $x = 0$ and $Q_2 = +1.0\,\mu C$ at $x = 20\,cm$.

(a) (3 points) Find the magnitude and direction of the electric field at the midpoint M ($x = 10\,cm$).

6 min



$$\vec{E}_{1x} = k\frac{Q_1}{r^2}\vec{e}_x = 9\cdot10^9\frac{N\,m^2}{C^2}\frac{9\cdot10^{-6}\,C}{(0.1\,m)^2}\vec{e}_x = 8.1\cdot10^6\frac{N}{C}\vec{e}_x$$

$$\vec{E}_{2x} = k\frac{Q_2}{r^2}(-\vec{e}_x) = 9\cdot10^9\frac{N\,m^2}{C^2}\frac{1\cdot10^{-6}\,C}{(0.1\,m)^2}(-\vec{e}_x) = 0.9\cdot10^6\frac{N}{C}(-\vec{e}_x)$$

$$\vec{E}_x = \vec{E}_{1x} + \vec{E}_{2x}$$

$$= \left(8.1\cdot10^6\frac{N}{C} - 0.9\cdot10^6\frac{N}{C}\right)\vec{e}_x$$

$$= 7.2\cdot10^6\frac{N}{C}\vec{e}_x$$

positive x-direction

too hard

15 min

(b) (3 points) At which point on the x-axis is the electric field zero?

$$\vec{E}_y = \vec{E}_{1y} + \vec{E}_{2y} = k\vec{e}_x\left(\frac{Q_1}{y^2} - \frac{Q_2}{(20cm-y)^2}\right) \overset{!}{=} \vec{0}$$

$$\Leftrightarrow \frac{Q_1}{y^2} - \frac{Q_2}{(20cm-y)^2} = 0 \qquad \frac{y}{20cm-y} = \sqrt{\frac{Q_1}{Q_2}} = 3$$

$$\frac{y^2}{(20cm-y)^2} = \frac{Q_1}{Q_2} \quad |\sqrt{\ }^*$$

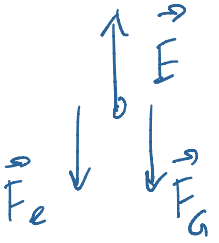
$$y = 60cm - 3y \Rightarrow y = 15\,cm$$

see also last page.

* Taking only the positive square root is ok, as both y and $(20cm - y)$ are expected to be positive in between the charges.

8 min

4. (5 points) An electron is in a uniform electric field of 500 N/C that points up. Calculate the electric force on the electron (magnitude and direction) and compare it with its weight. What is the acceleration of the electron?



$$F_E = E\cdot e = 500\frac{N}{C}\cdot 1.602\cdot10^{-19}\,C = 8.01\cdot10^{-17}\,N$$

$$F_G = g\cdot m_e = 9.81\frac{m}{s^2}\cdot 9.109\cdot10^{-31}\,kg = 8.936\cdot10^{-30}\,N$$

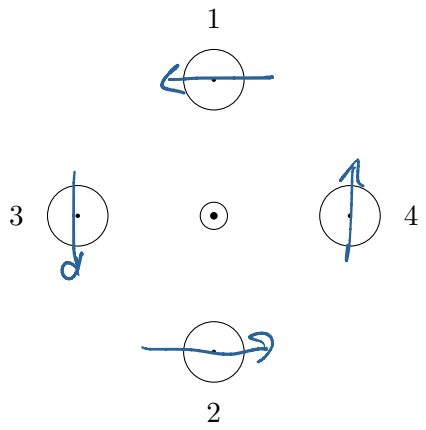
$$\vec{F}_{tot} = \vec{F}_E + \vec{F}_G \Rightarrow F_{tot} = F_E + F_G \approx F_E = 8.01\cdot10^{-17}\,N$$

$$F = m\cdot a \Rightarrow a = \frac{F}{m} = \frac{F_{tot}}{m_e} = \frac{8.01\cdot10^{-17}\,N}{9.109\cdot10^{-31}\,kg}$$

$$= 8.794\cdot10^{13}\frac{m}{s^2} \text{ downwards}$$

5. (4 points) **Compasses around a wire.** A long, straight wire is perpendicular to the page and carries a current *out of* the page. Four compasses are placed around it (neglect the Earth's magnetic field). Draw the direction in which the N pole of each compass points. What changes if the current is reversed?

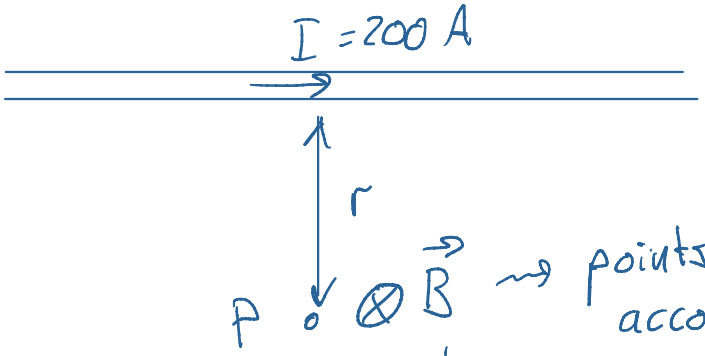
2 min



If the current is reversed, so is the magnetic field and thus the compass needles

6. (5 points) **Power line.** A long, horizontal power line in the plane of the page carries a current of 200 A to the right. (a) At what distance from the line is its magnetic field as strong as the Earth's magnetic field ($\approx 50\,\mu$T)? (b) In which direction does the line's field point at a point below the line?

7 min



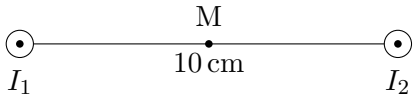
$\vec{B}$ ⇝ points into the page according to right-hand-rule

$$B_P = \frac{\mu_0}{2\pi}\frac{I}{r} \overset{!}{=} 50\,\mu T$$

$$r = \frac{\mu_0}{2\pi}\frac{I}{50\mu T} = \frac{2\cancel{4\pi}\cdot 10^{-7}\,\frac{Tm}{\cancel{A}}}{\cancel{2\pi}} \cdot \frac{200\,\cancel{A}}{50\cdot 10^{-6}\,T}$$

$$= \frac{4}{5}\,m = 0.8\,m$$

7. Two long, straight wires are perpendicular to the page and 10 cm apart. Both currents point *out of* the page: $I_1 = 30\,\text{A}$ in the left wire, $I_2 = 10\,\text{A}$ in the right wire.



(a) (4 points) Find the magnitude and direction of the magnetic field at the midpoint M.

7 min

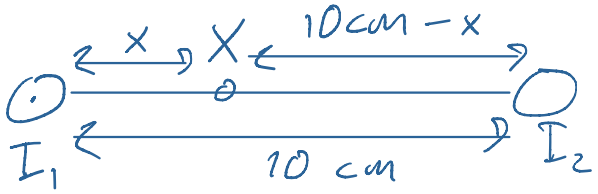
$$\vec{B}_M = \vec{B}_{M1} + \vec{B}_{M2} = (B_{M1} - B_{M2})\,\vec{e}_y$$

$$B_{M1} = \frac{\mu_0}{2\pi}\frac{I_1}{5\,\text{cm}} = \frac{4\pi\cdot 10^{-7}\,\frac{\text{Tm}}{\text{A}}}{2\pi}\,\frac{30\,\text{A}}{5\cdot 10^{-2}\,\text{m}} = \frac{6}{5}\cdot 10^{-4}\,\text{T} = 1.2\cdot 10^{-4}\,\text{T}$$

$$B_{M2} = \frac{\mu_0}{2\pi}\frac{I_2}{5\,\text{cm}} = \frac{\dots}{\dots}\,\frac{10\,\text{A}}{\dots} = \frac{B_{M1}}{3} = 0.4\cdot 10^{-4}\,\text{T}$$

$$\vec{B}_M = 0.8\cdot 10^{-4}\,\text{T}\;\vec{e}_y$$

4 min (b) (3 points) At which point between the wires is the magnetic field zero?



$$\vec{B}_X = (B_{X1} - B_{X2})\,\vec{e}_y = \vec{0}$$

$$B_{X1} = \frac{\mu_0}{2\pi}\frac{I_1}{x}, \quad B_{X2} = \frac{\mu_0}{2\pi}\frac{I_2}{10\,\text{cm} - x}$$

$$\frac{\mu_0}{2\pi}\left(\frac{I_1}{x} - \frac{I_2}{10\,\text{cm} - x}\right) \overset{!}{=} 0$$

$$(10\,\text{cm} - x)\,30\,\text{A} = x\cdot 10\,\text{A}$$

$$4x = 30\,\text{cm}$$

$$x = 7.5\,\text{cm}$$

Additional space for answers and notes

If you continue an answer here, write the question number next to it and add a note "continued on the last page" at the question.

Other solution of 3 b)

$$\vec{E}_y = \vec{E}_{1y} + \vec{E}_{2y} = k\,\vec{e}_x\left(\frac{Q_1}{y^2} - \frac{Q_2}{(20\,\text{cm}-y)^2}\right) \overset{!}{=} \vec{0}$$

$$\Leftrightarrow \frac{Q_1}{y^2} - \frac{Q_2}{(20\,\text{cm}-y)^2} = 0$$

$$\frac{9\,\cancel{\mu C}}{y^2} - \frac{1\,\cancel{\mu C}}{(20\,\text{cm}-y)^2} = 0 \;|\cdot y^2 \cdot (20\,\text{cm}-y)^2$$

$$9(20\,\text{cm}-y)^2 - y^2 = 0$$

$$8y^2 - 9\cdot 40\,\text{cm}\cdot y + 9\cdot 400\,\text{cm}^2 = 0$$

$$y_{1,2} = \frac{9\cdot 40 \pm \sqrt{9^2 40^2 - 4\cdot 8\cdot 9\cdot 400}}{2\cdot 8}\,\text{cm}$$

$$= \frac{360 \pm \sqrt{129'600 - 115'200}}{16}\,\text{cm}$$

$$= \frac{360 \pm 120}{16}\,\text{cm} = 30\,\text{cm},\ 15\,\text{cm}$$

Only $y = 15\,\text{cm}$ is a valid solution as $y = 30\,\text{cm}$ is not between the charges, and the superposition of the two field components would be with two plus signs, no minus, therefore yield a different expression that cannot be $\vec{0}$.